Cover letter prepared for Mathematics of Computation

27 October 2026

Dear Editors,

Please consider the manuscript *A sixteen-prime-factor bound for Lehmer's totient problem and exact prime-completion criteria*, by Deep Bhattacharjee. It gives a computational sixteen-prime-factor lower bound in Lehmer's totient problem, strengthened bounds in the stated quotient cases, and a finite classification for the companion divisibility with at most eight prime factors. The manuscript separates the written reductions, recorded exhaustive computations, independent certificates and partial formal verification.

This revision gives detailed exact prime-completion criteria, diagrams, and a proof that deletion of the largest prime cannot preserve a hypothetical Lehmer counterexample. The full Lehmer conjecture remains unproved. The submission therefore concerns the stated bounds and classifications, not a claimed resolution of that conjecture.

The original preprint is https://hexagonmath.org/2610.00454v1. The earlier confirmed research archive is https://doi.org/10.5281/zenodo.23638948. Code and verification records are available at https://github.com/creelie/eLhmr. AI assistance is disclosed in the manuscript and in the accompanying author-prepared declaration.

Yours sincerely,
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